Steam Power & Manufacturing

The Role of Steam Power in Economic Growth in 1850 – Variation of Effect Across States and Industries

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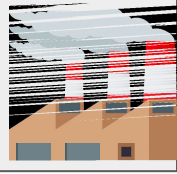


Background

Context

During the mid 19th century, the U.S. was transitioning from an economy based on agriculture to one that thrives off of manufacturing. Steam power was one of the most important advancements during this time, allowing manufacturing firms to rapidly scale their operational capacity. Steam power did not affect everyone evenly, though, certain regions and industries benefited much more than others. Our analysis draws from the 1850 Census of Manufacturing data, containing information about well over 100,000 firms across 33 states and 15 industries at the height of this transformation.

Research Question

To what extent did the transition to steam power drive the rapid increase in manufacturing productivity during this period, and how did it vary in different regions and industries?

Regression Model

Regression Equation

$$log(prod_val) = \beta_0 + \beta_1(steam) + \beta_2 log(capital) + \beta_3(state) + \beta_4(industry) + \beta_5(steam*industry) + \mu$$

Variable Definitions

- *prod_val* – total annual value of goods produced by each firm (in thousands)
- *steam* – dummy variable indicating whether the firm used steam power
- *capital* – total value invested in the firm including real estate, equipment, etc. (in thousands)
- *state* – state where the firm was primarily located
- *industry* – primary type of business (general categories)
- *steam*industry* – interaction effect between *steam* and *industry*, which displays the effects steam power had on each individual industry

References and AI Disclosure

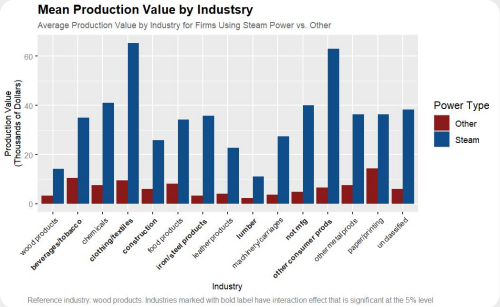
Hornbeck, R., Hsu, S. H.-M., Humlum, A., & Rotemberg, M. (2025). Gaining steam: Technology diffusion with recurring lock-in [Data set]. Historical Census of Manufactures Microdata. https://cmfdata.org

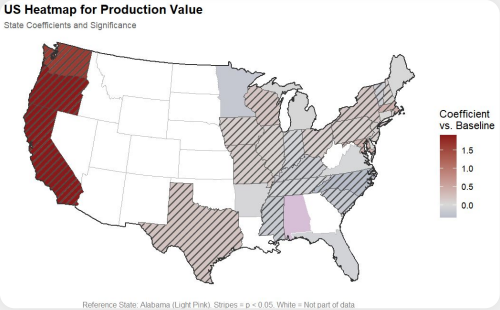
Claude Sonnet 4.6 was used for general planning and review when converting our analysis in R to this poster format. It helped with improving clarity regarding some of the interpretation of the code.
Gemini 3.1 was used due to the automatic AI overview provided when conducting a Google search. It helped answer some minor questions to ensure accuracy in our report. It was not directly prompted.
These AI tools were primarily just used for reviewing and editing the information we already had. All AI output was thoroughly reviewed before being taken into consideration or implemented into the project. The authors of this project assume full responsibility for the accuracy and proper use of all AI use regarding this project. More details can be provided upon request.

Data and Variables

Statistic	N	Mean	St. Dev.	Min	Max
capital	124,860	4.3984	27.8071	0	3,000
prod_val	124,860	7.4673	47.1090	0	7,458.58
steam	124,860	0.0677	0.2513	0	1

Note: *capital* and *prod_val* are expressed in thousands of dollars





Discussion

Results Table

Independent Variable	Estimate	St. Error
lcapital	0.574***	(0.002)
steam	0.452***	(0.043)
state	some significant at 5% (see graph)	
industry	all significant at 5%	
steam:industry	some significant at 5% (see graph)	
R2	0.516	
Adjusted R2	0.516	
Residual Std. Error	0.923	(df = 124,797)
F Statistic	2,148.09***	(df = 62; 124,797)

Note: * p < 0.1; ** p < 0.05; *** p < 0.01

Understanding the Results

We see that *steam* and *log(capital)* are both significant at the 1% level. Holding all else constant, firms using steam power produced almost 46% greater than their counterparts. The strongest predictor appears to be *log(capital)*, where a 1% increase corresponds to a 57.4% increase in production value. Steam power had a significant interaction with half of the included industries, which makes sense as some industries were already high producing. Geography also played an interesting role. We see that the west coast states were significant and had large positive coefficients. This reflects the rapid economic expansion driven by the gold rush.

Steam Power's Role in Industrialization

The regression results indicate that steam power had a statistically significant effect on production output in 1850, with close to a 46% increase in production value. Since capital investment displayed an even greater effect, this indicates that financing was just as important as the technological advancements. In addition, we see that steam power was far more important in certain industries than others, so it undoubtedly played a large role in increased production, but the effect was far from uniform.